Homework 9

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Problem 9.1.

Solution 9.1.

1. Set $H = \begin{bmatrix} \boldsymbol{h}_1 & \boldsymbol{h}_2 & \cdots & \boldsymbol{h}_n \end{bmatrix}$, where the columns are mutually perpendicular and all entries are ± 1, then

$$H^{\mathrm{T}} H = \begin{bmatrix} \boldsymbol{h}_1^{\mathrm{T}} \\ \boldsymbol{h}_2^{\mathrm{T}} \\ \vdots \\ \boldsymbol{h}_n^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{h}_1 & \boldsymbol{h}_2 & \cdots & \boldsymbol{h}_n \end{bmatrix} = nI_n.$$

2. $\begin{bmatrix} H^{\mathrm{T}} & H^{\mathrm{T}} \\ H^{\mathrm{T}} & -H^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} H & H \\ H & -H \end{bmatrix} = \begin{bmatrix} 2nI_n & \\ & 2nI_n \end{bmatrix} = 2nI_{2n}$. Each columns of the new matrix are perpendicular to each other, and their entries are ± 1, so it is Hadamard matrix.

3. Assumed that there exists a 3×3 Hadamard matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then $a_{11}a_{12} + a_{21}a_{22} + a_{31}a_{32} = 0$. But $a_{11}a_{12}, a_{21}a_{22}, a_{31}a_{32}$ are all ± 1. Contradiction. So the 3×3 Hadamard matrix doesn't exist.

4.
Proof:
Set the row operations for permutation is P_1, the column operations for permutation is P_2. Then $P_1 = \begin{bmatrix} \boldsymbol{e}_{i_1} & \boldsymbol{e}_{i_2} & \cdots & \boldsymbol{e}_{i_n} \end{bmatrix}$, where $i_1, i_2, \cdots, i_n$ are $1, 2, \cdots, n$ in an arbitrary order. So $P_1^{\mathrm{T}} P_1 = I_n$. Similarly, $P_2^{\mathrm{T}} P_2 = I_n$. Set the row operations for negation (scaling) is Q_1, the column operations for negation (scaling) is Q_2. Then $Q_1 = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix}$, where $d_1, d_2, \cdots, d_n = \pm 1$. So $Q_1^{\mathrm{T}} Q_1 = I_n$. Similarly, $Q_2^{\mathrm{T}} Q_2 = I_n$.
After permutation and negation, the new matrix is $Q_1 P_1 H P_2 Q_2$. Thus,

$$(Q_1 P_1 H P_2 Q_2)^{\mathrm{T}} (Q_1 P_1 H P_2 Q_2) = Q_2^{\mathrm{T}} P_2^{\mathrm{T}} H^{\mathrm{T}} P_1^{\mathrm{T}} Q_1^{\mathrm{T}} Q_1 P_1 H P_2 Q_2 = nI_n$$

Hence each column of $Q_1 P_1 H P_2 Q_2$ is still mutually orthogonal and all entries are $\pm 1 \implies$ the result is still Hadamard matrix.

5.
Proof:
By permutation and negation, any 4×4 Hadamard matrix can be converted to $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & * & * & * \\ 1 & * & * & * \\ 1 & * & * & * \end{bmatrix}$, where $*$ is an undetermined entry. By the definition of Hadamard matrix, the three $*$ in the second column the and second row must be two -1 and one 1 in an arbitrary order. We continue permuting the last three columns and rows (we can not negate now) until the matrix becomes $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & * & * \\ 1 & -1 & * & * \end{bmatrix}$. Again, the two undetermined entries in the third column must be a 1 and a -1 in an arbitrary order. Then, we get $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & * \\ 1 & -1 & -1 & * \end{bmatrix}$ by row permutation. Now the last two $*$ are determined, the matrix is $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$.

That is to say, any 4×4 Hadamard matrix can be converted to $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$ by row/column negations and permutations. So they are equivalent.

Problem 9.2.

Solution 9.2.

1. By the definition of Gram Matrix, $D = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{bmatrix}$. It's symmetric but not positive definite.

2. Proof:
The "inner product" here is symmetric and bilinear, but not positive definite. So $\boldsymbol{v} \perp W \iff \boldsymbol{v}$ perpendicular to all the basis of W. Set the basis of W is $\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_r$, and $A = \begin{bmatrix} \boldsymbol{w}_1 & \boldsymbol{w}_2 & \cdots & \boldsymbol{w}_r \end{bmatrix}$, then $\dim W = r$. From 1, $\boldsymbol{v} \perp \boldsymbol{w}$ if and only if $\boldsymbol{w}^{\mathrm{T}} D\boldsymbol{v} = 0$, therefore,

$$W^{\perp} = \{\boldsymbol{v} \mid \boldsymbol{v} \perp W\} = \{\boldsymbol{v} \mid \boldsymbol{v} \perp \boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_r\} = \left\{\boldsymbol{v} \mid \boldsymbol{w}_1^{\mathrm{T}} D\boldsymbol{v} = 0, \boldsymbol{w}_2^{\mathrm{T}} D\boldsymbol{v} = 0, \cdots, \boldsymbol{w}_r^{\mathrm{T}} D\boldsymbol{v} = 0\right\} = \ker(A^{\mathrm{T}} D)$$

By rank-nullity Theorem, we have

$$\dim W^{\perp} = \dim \ker(A^{\mathrm{T}} D) = 4 - \dim \operatorname{Ran}(A^{\mathrm{T}} D) = 4 - \operatorname{rank}(A^{\mathrm{T}} D) = 4 - \operatorname{rank}(A^{\mathrm{T}}) = 4 - r$$

Hence $\dim W + \dim W^{\perp} = 4$.
3. $\mathbb{R}^4$. The orthogonal complement of $\mathbb{R}^4$ is $\{\mathbf{0}\}$, and $\{\mathbf{0}\} \subset \mathbb{R}^4$.
4. By observation, the length of $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \in \mathbb{R}^4$ are zero, and they are linearly independent. So they are basis of V.
Problem 9.3.

Solution 9.3.

1. $A = \begin{bmatrix} 2 & 2 & \\ 2 & 4 & 2 \\ & 2 & 3 \end{bmatrix}$, and the LDU decomposition of A is

$$A = LDU = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 2 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \\ & 1 & 1 \\ & & 1 \end{bmatrix}$$

Since the diagonal entries of D are all positive, the matrix are positive definite.
The Cholesky decomposition of A is

$$\begin{aligned} A &= \begin{bmatrix} 1 & & \\ 1 & 1 & \\ & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 2 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \\ & 1 & 1 \\ & & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & & \\ 1 & 1 & \\ & 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{2} & \\ & & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{2} & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \\ & 1 & 1 \\ & & 1 \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2} & & \\ \sqrt{2} & \sqrt{2} & \\ & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} & \\ & \sqrt{2} & \sqrt{2} \\ & & 1 \end{bmatrix} \end{aligned}$$

And $f(x, y, z) = (\sqrt{2}x + \sqrt{2}y)^2 + (\sqrt{2}y + \sqrt{2}z)^2 + z^2$.

2. $A = \begin{bmatrix} 4 & 2 & \\ 2 & 2 & 1 \\ & 1 & 1 \end{bmatrix}$, and the LU decomposition of A is

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is not positive definite because it's not invertible.

Problem 9.4.

Solution 9.4.

1.
Since $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ is orthonormal basis, $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ are mutually perpendicular and their length is 1.

$$\begin{aligned} \boldsymbol{b}_1^{\mathrm{T}} \boldsymbol{b}_2 &= \frac{1}{9}(2\boldsymbol{a}_1 + 2\boldsymbol{a}_2 - \boldsymbol{a}_3)^{\mathrm{T}}(2\boldsymbol{a}_1 - \boldsymbol{a}_2 + 2\boldsymbol{a}_3) \\ &= \frac{1}{9}(4\boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{a}_1 - 2\boldsymbol{a}_2^{\mathrm{T}}\boldsymbol{a}_2 - \boldsymbol{a}_3^{\mathrm{T}}\boldsymbol{a}_3) \\ &= \frac{1}{9}(4 - 2 - 2) \\ &= 0 \end{aligned}$$

Similarly, $\boldsymbol{b}_1^{\mathrm{T}}\boldsymbol{b}_3 = 0, \boldsymbol{b}_2^{\mathrm{T}}\boldsymbol{b}_3 = 0$. So they are mutually perpendicular. Besides, $\boldsymbol{b}_1^{\mathrm{T}}\boldsymbol{b}_1 = 1, \boldsymbol{b}_2^{\mathrm{T}}\boldsymbol{b}_2 = 1, \boldsymbol{b}_3^{\mathrm{T}}\boldsymbol{b}_3 = 1$, so $\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3$ is orthonormal basis.

2. They are not perpendicular, so let's apply Gram-Schmidt Orthogonalization to them.
Starting from $\boldsymbol{b}_1$, we do dot product with all the vectors with $\boldsymbol{b}_1$.

	$\boldsymbol{a}_1 + \boldsymbol{a}_5$	$\boldsymbol{a}_1 - \boldsymbol{a}_2 + \boldsymbol{a}_4$	$2\boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3$
dot product with $\boldsymbol{b}_1$	2	1	2

Then we use the first column to kill the dot product by column operations.

$$\hat{\boldsymbol{q}}_1 = \boldsymbol{b}_1 = \boldsymbol{a}_1 + \boldsymbol{a}_5$$

$$\hat{\boldsymbol{q}}_2 = \boldsymbol{b}_2 - \frac{1}{2}\boldsymbol{b}_1 = \frac{1}{2}\boldsymbol{a}_1 - \boldsymbol{a}_2 + \boldsymbol{a}_4 - \frac{1}{2}\boldsymbol{a}_5$$

$$\hat{\boldsymbol{q}}_3 = \boldsymbol{b}_3 - \boldsymbol{b}_1 = \boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3 - \boldsymbol{a}_5$$

Now $\hat{\boldsymbol{q}}_1$ and $\hat{\boldsymbol{q}}_2, \hat{\boldsymbol{q}}_3$ are perpendicular. Coincidently, $\hat{\boldsymbol{q}}_2$ and $\hat{\boldsymbol{q}}_3$ are perpendicular too. Then, we need to scale them such that their length is 1.

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(\boldsymbol{a}_1 + \boldsymbol{a}_5)$$

$$\boldsymbol{q}_2 = \frac{\sqrt{10}}{5}(\frac{1}{2}\boldsymbol{a}_1 - \boldsymbol{a}_2 + \boldsymbol{a}_4 - \frac{1}{2}\boldsymbol{a}_5)$$

$$\boldsymbol{q}_3 = \frac{1}{2}(\boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3 - \boldsymbol{a}_5)$$

And they are orthonormal basis now.

Problem 9.5.

Solution 9.5.

1.

$V = \ker(\begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix})$, by the Fundamental Theorem of Linear Algebra, $V^{\perp} = \ker(\begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix})^{\perp} = \mathrm{Ran}(\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix})$.

Then the basis of $V^{\perp}$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, so the orthonormal basis (ONB) of $V^{\perp}$ is $\boldsymbol{q}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$. A basis of V is apparently $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.

The orthogonal projection matrix to $V^{\perp}$ is

$$P = \boldsymbol{q}_1 \boldsymbol{q}_1^{\mathrm{T}} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$$

So $\boldsymbol{b}_2 = P\boldsymbol{b} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$, and $\boldsymbol{b}_1 = \boldsymbol{b} - \boldsymbol{b}_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$. Then

$$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

2.
$V = \mathrm{span}(\boldsymbol{e}_1, \boldsymbol{e}_1 + \boldsymbol{e}_2) = \mathrm{span}(\boldsymbol{e}_1, \boldsymbol{e}_2)$, so the ONB of V is $\boldsymbol{e}_1, \boldsymbol{e}_2$, then obviously the ONB of $V^{\perp}$ is $\boldsymbol{e}_3, \boldsymbol{e}_4$. Hence,

$$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \\ 7 \end{bmatrix}$$

3.
A basis of V is $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, then the ONB of V is $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$. A basis of $V^{\perp}$ is $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$. Set $Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{bmatrix}$, the orthogonal projection matrix to V is

$$P = QQ^{\mathrm{T}} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

So $\boldsymbol{b}_1 = P\boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{b}_1 = \begin{bmatrix} 3 \\ -1 \\ -2 \\ 0 \end{bmatrix}$. Thus,

$$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ -2 \\ 0 \end{bmatrix}.$$

Problem 9.6.

Solution 9.6.

1.

$$\frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a & -c \\ c & a \end{bmatrix}^{\mathrm{T}} \frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a & -c \\ c & a \end{bmatrix} = I_2$$

So Q is orthogonal matrix. And $\frac{1}{\sqrt{a^2+c^2}}\begin{bmatrix} a^2+c^2 & ab+cd \\ 0 & ad-bc \end{bmatrix}$ is an upper triangular matrix. Furthermore,

$$QR = \frac{1}{\sqrt{a^2+c^2}}\begin{bmatrix} a & -c \\ c & a \end{bmatrix}\frac{1}{\sqrt{a^2+c^2}}\begin{bmatrix} a^2+c^2 & ab+cd \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A.$$

So they are QR decomposition of A.

2.

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & 3 \\ & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$$

3. In QR decomposition, the first column doesn't change except normalization. Let's say the first column of A is $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, then the first column of Q is $\frac{1}{\sqrt{a^2+b^2+c^2}}\begin{bmatrix} a \\ b \\ c \end{bmatrix}$.